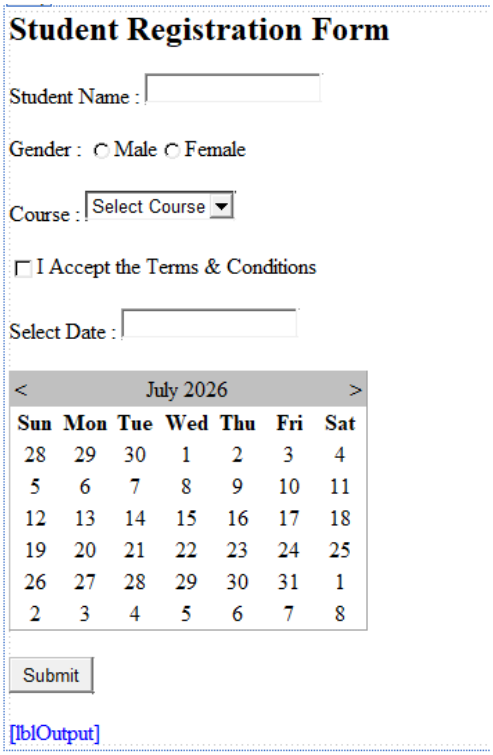
4. Create an ASP.NET Web Forms application using simple server controls.

4(a). Create an ASP.NET Web Forms application using simple server controls. use c#. Design as a image. a. Design a Student Registration Form using the following ASP.NET controls:

i. Label and TextBox to accept Student Name
ii. RadioButton to select Gender (Male/Female)
iii. DropDownList to select Course (BSc IT, BSc CS, BCA)
iv. CheckBox for accepting Terms & Conditions
v. Calendar control to select Date
vi. Button control to submit the form
vii. Label control to display the output.



Solution:

.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeFile="WebForm1.aspx.cs" Inherits="WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
  <title>Student Registration Form</title>
</head>
<body>

<form id="form1" runat="server">
  <h2>Student Registration Form</h2>
  <asp:Label ID="lblName" runat="server" Text="Student Name : "></asp:Label>
  <asp:TextBox ID="txtName" runat="server"></asp:TextBox>
  <br /><br />
  <asp:Label ID="lblGender" runat="server" Text="Gender : "></asp:Label>
  <asp:RadioButton ID="rbMale" runat="server"  GroupName="Gender"  Text="Male" />
  <asp:RadioButton ID="rbFemale" runat="server"  GroupName="Gender"  Text="Female" />
  <br /><br />
  <asp:Label ID="lblCourse" runat="server" Text="Course : "></asp:Label>
  <asp:DropDownList ID="ddlCourse" runat="server">
    <asp:ListItem>Select Course</asp:ListItem>
    <asp:ListItem>BSc IT</asp:ListItem>
    <asp:ListItem>BSc CS</asp:ListItem>
    <asp:ListItem>BCA</asp:ListItem>
  </asp:DropDownList>
  <br /><br />
  <asp:CheckBox ID="chkTerms" runat="server" Text="I Accept the Terms & Conditions" />
  <br /><br />
  <asp:Label ID="lblDate" runat="server" Text="Select Date : ">  </asp:Label>
  <asp:TextBox ID="txtDate" runat="server" ReadOnly="true"> </asp:TextBox>
  <br /><br />
  <asp:Calendar ID="Calendar1" runat="server" OnSelectionChanged="Calendar1_SelectionChanged">
  </asp:Calendar>
  <br />
  <asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
  <br /><br />
  <asp:Label ID="lblOutput" runat="server" ForeColor="Blue"> </asp:Label>
</form>
</body>
</html>
```

Aspx.cs

```
using System;

public partial class WebForm1 : System.Web.UI.Page
{
    //Display selected date in TextBox
    protected void Calendar1_SelectionChanged(object sender, EventArgs e)
    {
        txtDate.Text = Calendar1.SelectedDate.ToShortDateString();
    }

    //Submit button
    protected void btnSubmit_Click(object sender, EventArgs e)
    {
        string gender = "";

        if (rbMale.Checked)
            gender = "Male";
        else if (rbFemale.Checked)
            gender = "Female";

        string terms;

        if (chkTerms.Checked)
            terms = "Accepted";
        else
            terms = "Not Accepted";

        lblOutput.Text = "<b>Registration Details</b><br/><br/>";
        lblOutput.Text += "Student Name : " + txtName.Text + "<br/>";
        lblOutput.Text += "Gender : " + gender + "<br/>";
        lblOutput.Text += "Course : " + ddlCourse.SelectedItem.Text + "<br/>";
        lblOutput.Text += "Date : " + txtDate.Text + "<br/>";
        lblOutput.Text += "Terms : " + terms;
    }
}
```

4(b). Registration Form using Validation Controls Validation Controls Used

- **RequiredFieldValidator** – Checks that the user does not leave a required field empty.
- **CompareValidator** – Compares the value of one field with another field or a fixed value.
- **RangeValidator** – Checks that the entered value is within a specified range.
- **RegularExpressionValidator** – Checks that the input matches a specific pattern, such as an email or phone number.
- **ValidationSummary** – Displays all validation error messages in one place.

Solution:

.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication1.WebForm1" %>
<!DOCTYPE html>
<html>
<head runat="server">
  <title>User Registration Form</title>
  <style>
    body{
      font-family:Arial;
      background:#f4f4f4;
    }
    .container{
      width:900px;
      margin:auto;
      border:1px solid #999;
      padding:10px;
      background:white;
    }
    h2{
      background:#2f5fb3;
      color:white;
      text-align:center;
      padding:10px;
      margin-top:0;
    }
    table{
      width:100%;
    }
    td{
      padding:6px;
      vertical-align:top;
    }
    .textbox{
      width:300px;
      padding:5px;
    }
    .error{
      color:red;
      font-size:14px;
    }
    .success{
      color:green;
      font-weight:bold;
    }
  </style>
</head>
<body>
```



```
<form id="form1" runat="server">
<div class="container">
<h2>User Registration Form</h2>
<table>
<tr>
<td>Full Name *</td>
<td>
<asp:TextBox ID="txtName" runat="server"
 CssClass="textbox"></asp:TextBox>
</td>
<td>
<asp:RequiredFieldValidator ID="rfvName"
runat="server" ControlToValidate="txtName"
ErrorMessage="Please enter your full name."
CssClass="error">
</asp:RequiredFieldValidator>
</td>
</tr>

<tr>
<td>User Name *</td>
<td>
<asp:TextBox ID="txtUser" runat="server" CssClass="textbox"></asp:TextBox>
</td>
<td>
<asp:RequiredFieldValidator ID="rfvUser" runat="server" ControlToValidate="txtUser"
ErrorMessage="User name is required." CssClass="error">
</asp:RequiredFieldValidator>
</td>
</tr>

<tr>
<td>Password *</td>
<td>
<asp:TextBox ID="txtPassword" runat="server" TextMode="Password" CssClass="textbox">
</asp:TextBox>
</td>
<td>
<asp:RegularExpressionValidator ID="revPassword" runat="server" ControlToValidate="txtPassword"
ValidationExpression=".{6,}" ErrorMessage="Password must be at least 6 characters." CssClass="error">
</asp:RegularExpressionValidator>
</td>
</tr>

<tr>
<td>Confirm Password *</td>
<td>
<asp:TextBox ID="txtConfirm" runat="server" TextMode="Password" CssClass="textbox">
</asp:TextBox>
</td>
<td>
<asp:CompareValidator ID="cvPassword" runat="server" ControlToCompare="txtPassword"
ControlToValidate="txtConfirm" ErrorMessage="Passwords do not match." CssClass="error">
</asp:CompareValidator>
</td>
</tr>
```



```
<tr>
<td>Email *</td>
<td>
<asp:TextBox ID="txtEmail" runat="server" CssClass="textbox"></asp:TextBox>
</td>
<td>
<asp:RegularExpressionValidator ID="revEmail" runat="server" ControlToValidate="txtEmail"
ValidationExpression="\w+([-.']\w+)*@\w+([-.]\w+)*\.\w+([-.]\w+)*" ErrorMessage="Please enter a
valid email address." CssClass="error">
</asp:RegularExpressionValidator>
</td>
</tr>

<tr>
<td>Mobile No *</td>
<td>
<asp:TextBox ID="txtMobile" runat="server" CssClass="textbox"></asp:TextBox>
</td>
<td>
<asp:RegularExpressionValidator ID="revMobile" runat="server" ControlToValidate="txtMobile"
ValidationExpression="^[0-9]{10}$" ErrorMessage="Enter a valid 10 digit mobile number."
CssClass="error">
</asp:RegularExpressionValidator>
</td>
</tr>
<tr>
<td>Date of Birth *</td>
<td>
<asp:TextBox ID="txtDOB" runat="server" TextMode="Date" CssClass="textbox"></asp:TextBox>
</td>
<td>
<asp:RequiredFieldValidator ID="rfvDOB" runat="server" ControlToValidate="txtDOB"
ErrorMessage="Please select your date of birth." CssClass="error">
</asp:RequiredFieldValidator>
</td>
</tr>

<tr>
<td>Gender *</td>
<td>
<asp:RadioButtonList ID="rblGender" runat="server" RepeatDirection="Horizontal">
<asp:ListItem>Male</asp:ListItem>
<asp:ListItem>Female</asp:ListItem>
</asp:RadioButtonList>
</td>
<td>
<asp:RequiredFieldValidator  ID="rfvGender" runat="server" ControlToValidate="rblGender"
InitialValue="" ErrorMessage="Please select your gender." CssClass="error">
</asp:RequiredFieldValidator>
</td>
</tr>

<tr>
<td>Country *</td>
<td>
```

```
<asp:DropDownList ID="ddlCountry" runat="server" Width="310px">
<asp:ListItem Value="">-- Select Country --</asp:ListItem>
<asp:ListItem>India</asp:ListItem>
<asp:ListItem>USA</asp:ListItem>
<asp:ListItem>UK</asp:ListItem>
<asp:ListItem>Canada</asp:ListItem>
</asp:DropDownList>
</td>
<td>
<asp:RequiredFieldValidator  ID="rfvCountry" runat="server" ControlToValidate="ddlCountry"
InitialValue="" ErrorMessage="Please select your country." CssClass="error">
</asp:RequiredFieldValidator>
</td>
</tr>

<tr>
<td colspan="2">
<asp:CheckBox ID="chkTerms" runat="server" Text="I accept the Terms &amp;amp; Conditions" />
</td>

<td>
<asp:CustomValidator ID="cvTerms" runat="server" ErrorMessage="You must accept the Terms &amp;amp;
Conditions." CssClass="error" ClientValidationFunction="ValidateTerms">
</asp:CustomValidator>
</td>
</tr>

<tr>
<td></td>
<td>
<asp:Button ID="btnRegister" runat="server" Text="Register" />
<asp:Button ID="btnReset" runat="server" Text="Reset" CausesValidation="false"/>
</td>
</tr>
<tr>
<td colspan="3">
<asp:ValidationSummary ID="ValidationSummary1" runat="server" ForeColor="Red" />
<br />

<asp:Label ID="lblMessage" runat="server" CssClass="success" Text="Registration Successful! Thank
you for registering.">
</asp:Label>
</td>
</tr>
</table>
</div>
</form>

</body>
</html>
```

ValidationExpression=".{6,}": . → Matches any single character, {6,} → Means **6 or more times**.
ValidationExpression="^[0-9]{10}$": ^ = start, [0-9]{10} = exactly 10 digits, $ = end of the input.

4(c). Create Web Form to demonstrate use of Adrotator Control.
Solution:

What is AdRotator Control?

Answer: The **AdRotator** control is an ASP.NET server control used to display advertisements randomly from an XML file. It automatically rotates ads each time the page is loaded. It allows users to click the advertisement and navigate to the specified website.

Solution:

```
<%@ Page Language="C#" AutoEventWireup="true" %>

<!DOCTYPE html>

<html>
<head runat="server">
   <title>AdRotator Control Demo</title>
</head>

<body>
<form id="form1" runat="server">
   <h2>Advertisement Rotator Demo</h2>
   <asp:AdRotator ID="AdRotator1"
     runat="server"
     AdvertisementFile="Ads.xml" Width="400px"
    Height="200px" Target="_blank">
   </asp:AdRotator>
</form>
</body>
</html>
```

Ads.xml

```
<?xml version="1.0" encoding="utf-8" ?>

<Advertisements>
 <Ad>
  <ImageUrl>Images/laptop.jpg</ImageUrl>
  <NavigateUrl>https://www.hp.com</NavigateUrl>
  <AlternateText>HP Laptop</AlternateText>
  <Impressions>40</Impressions>
 </Ad>

 <Ad>
  <ImageUrl>Images/mobile.jpg</ImageUrl>
  <NavigateUrl>https://www.samsung.com</NavigateUrl>
  <AlternateText>Samsung Mobile</AlternateText>
  <Impressions>30</Impressions>
 </Ad>

 <Ad>
  <ImageUrl>Images/headphone.jpg</ImageUrl>
  <NavigateUrl>https://www.boat-
lifestyle.com</NavigateUrl>
  <AlternateText>Boat Headphones</AlternateText>
  <Impressions>30</Impressions>
 </Ad>
</Advertisements>
```

